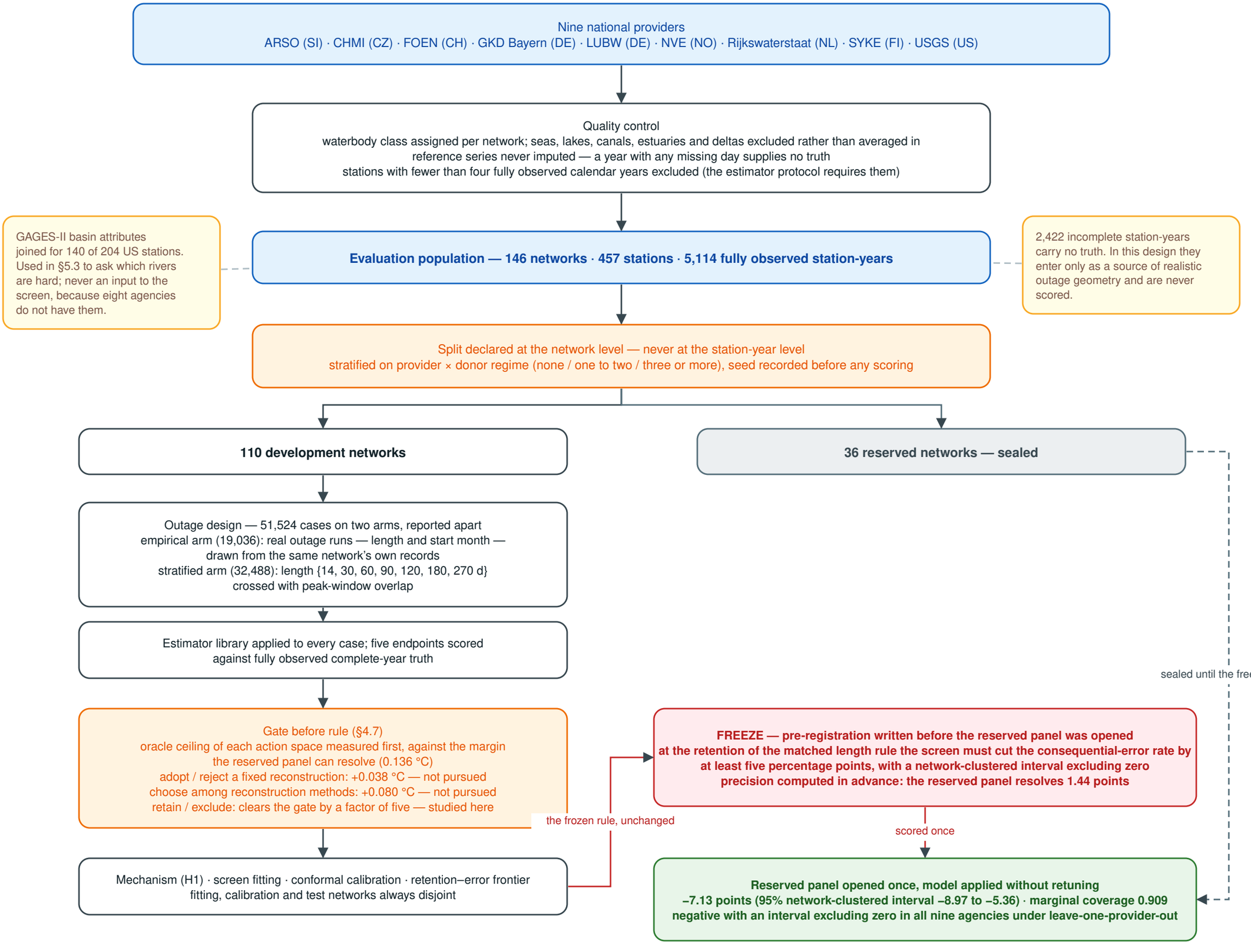
Nine national providers
ARSO (SI) · CHMI (CZ) · FOEN (CH) · GKD Bayern (DE) · LUBW (DE) · NVE (NO) · Rijkswaterstaat (NL) · SYKE (FI) · USGS (US)
Quality control
waterbody class assigned per network; seas, lakes, canals, estuaries and deltas excluded rather than averaged in
reference series never imputed — a year with any missing day supplies no truth
stations with fewer than four fully observed calendar years excluded (the estimator protocol requires them)
GAGES-II basin attributes
joined for 140 of 204 US stations.
Used in §5.3 to ask which rivers
are hard; never an input to the
screen, because eight agencies
do not have them.
Evaluation population — 146 networks · 457 stations · 5,114 fully observed station-years
2,422 incomplete station-years
carry no truth. In this design they
enter only as a source of realistic
outage geometry and are never
scored.
Split declared at the network level — never at the station-year level
stratified on provider × donor regime (none / one to two / three or more), seed recorded before any scoring
110 development networks
36 reserved networks — sealed
Outage design — 51,524 cases on two arms, reported apart
empirical arm (19,036): real outage runs — length and start month —
drawn from the same network's own records
stratified arm (32,488): length {14, 30, 60, 90, 120, 180, 270 d}
crossed with peak-window overlap
Estimator library applied to every case; five endpoints scored
against fully observed complete-year truth
Gate before rule (§4.7)
oracle ceiling of each action space measured first, against the margin
the reserved panel can resolve (0.136 °C)
adopt / reject a fixed reconstruction: +0.038 °C — not pursued
choose among reconstruction methods: +0.080 °C — not pursued
retain / exclude: clears the gate by a factor of five — studied here
Mechanism (H1) · screen fitting · conformal calibration · retention–error frontier
fitting, calibration and test networks always disjoint
the frozen rule, unchanged
FREEZE — pre-registration written before the reserved panel was opened
at the retention of the matched length rule the screen must cut the consequential-error rate by
at least five percentage points, with a network-clustered interval excluding zero
precision computed in advance: the reserved panel resolves 1.44 points
scored once
Reserved panel opened once, model applied without retuning
–7.13 points (95% network-clustered interval –8.97 to –5.36) · marginal coverage 0.909
negative with an interval excluding zero in all nine agencies under leave-one-provider-out
sealed until the freeze